AI BASED DEMAND FORECASTING INTELLIGENCE SYSTEM

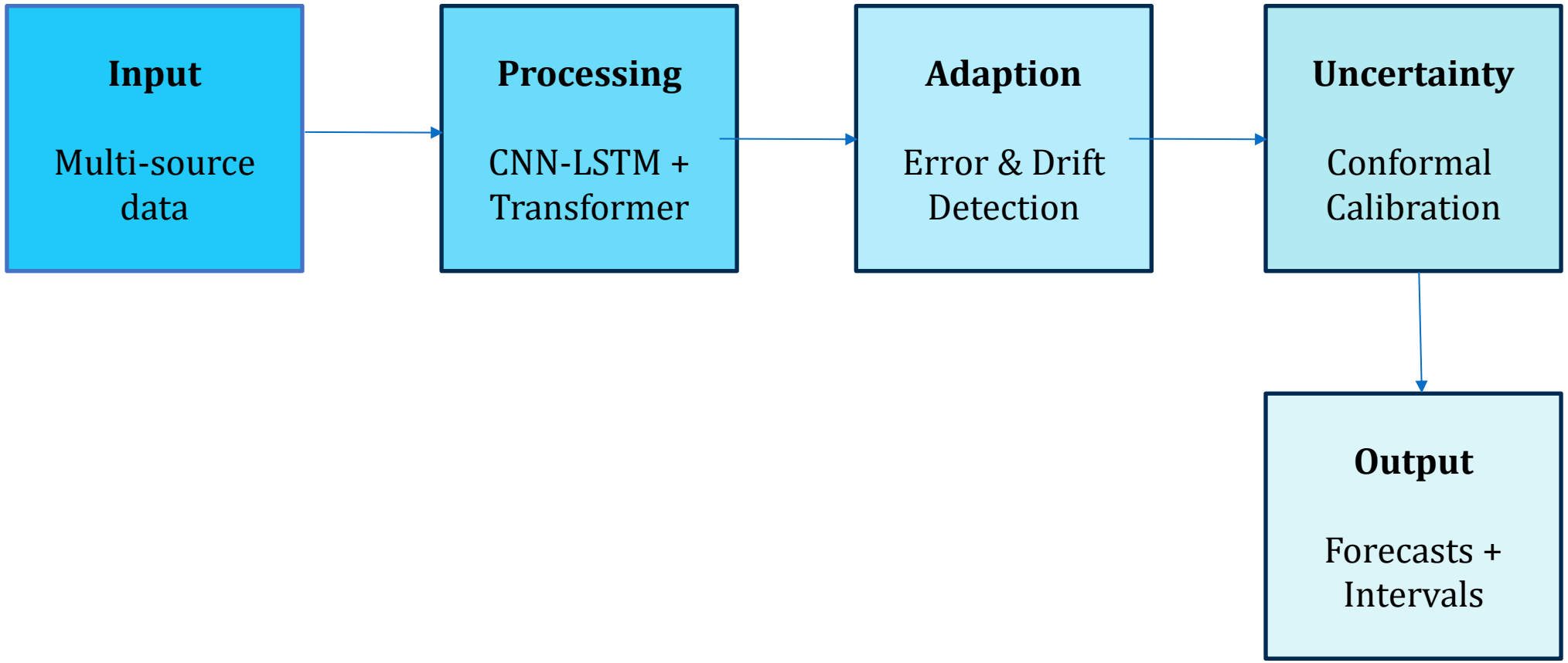
Component : Adaptive Forecasting and PredictionSystem

Comments: The scope of the component is too vague

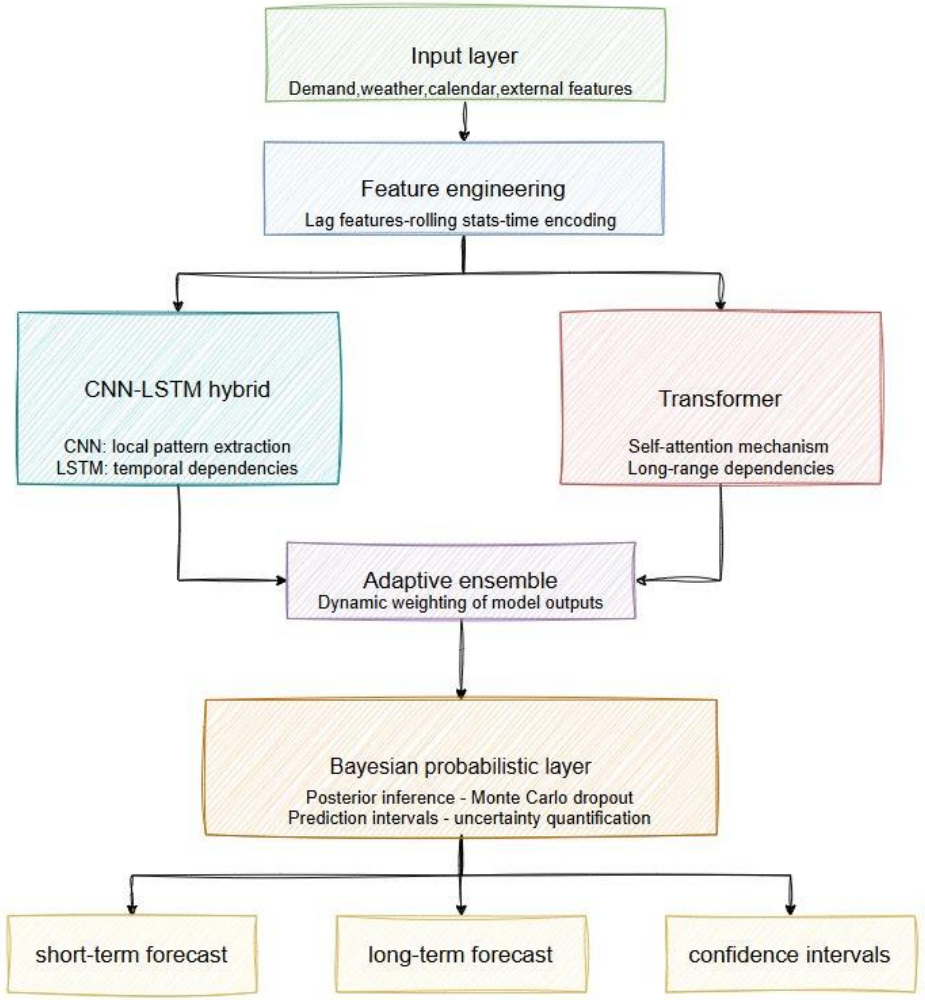
Purpose - Generate electricity forecasts with probabilistic uncertainty.

Key Differentiators
Multisource data integration
Conformal calibration for standard coverage
Real-time drift detection

Clear Component Pipeline



Component Architecture Diagram



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IT22069740 | JANAHAN .R

Adaptive Forecasting and Prediction System

Specialization: Information Technology

Adaptive Forecasting and Prediction System

User Requirements

Energy Planners & Analysts

- Demand patterns, long-term forecasts with confidence intervals

Utility Companies (CEB)

- Prevent blackouts, minimize waste , grid stability , interpretable forecasts

Industrial/Commercial Users

- Understand impact of their consumption on grid, optimize usage

Adaptive Forecasting and Prediction System

The Sub Problem:

Existing LSTM based models give only point forecasts (single value) with no uncertainty info.
Grid operators need confidence intervals to plan for worst-case vs best-case scenarios

Our Solution:

- Conformal Calibration Layer
- Multi-source Data Integration
- Uncertainty Quantification

Adaptive Forecasting and Prediction System

Design Excellence

Novelty

First hybrid CNN-LSTM + Transformer ensemble for Sri Lankan electricity forecasting

Uncertainty

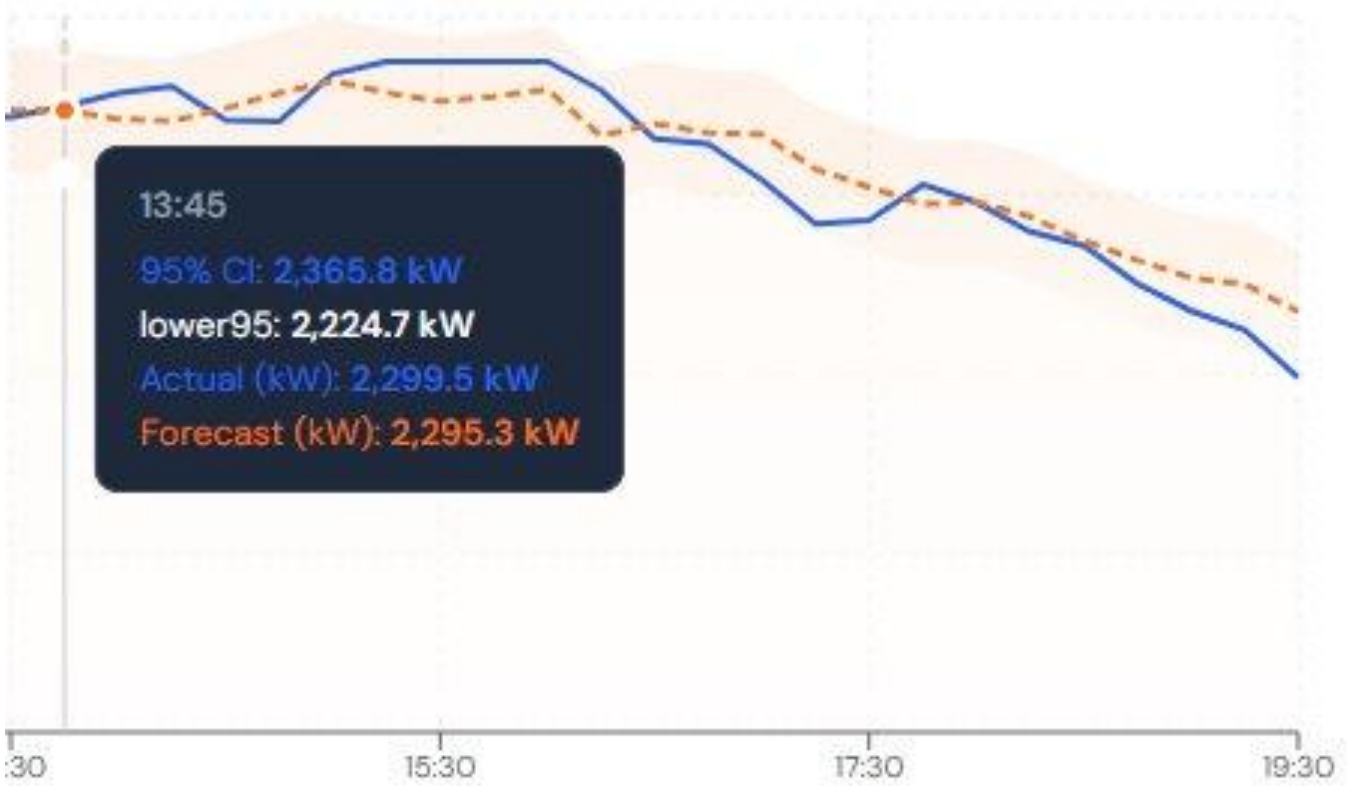
Guarantees 95% coverage- 95% of actuals MUST fall within predicted intervals

Adaptability

Dynamic model weighting reacts to concept drift – not static like traditional ensembles

Adaptive Forecasting and Prediction System

Prototype



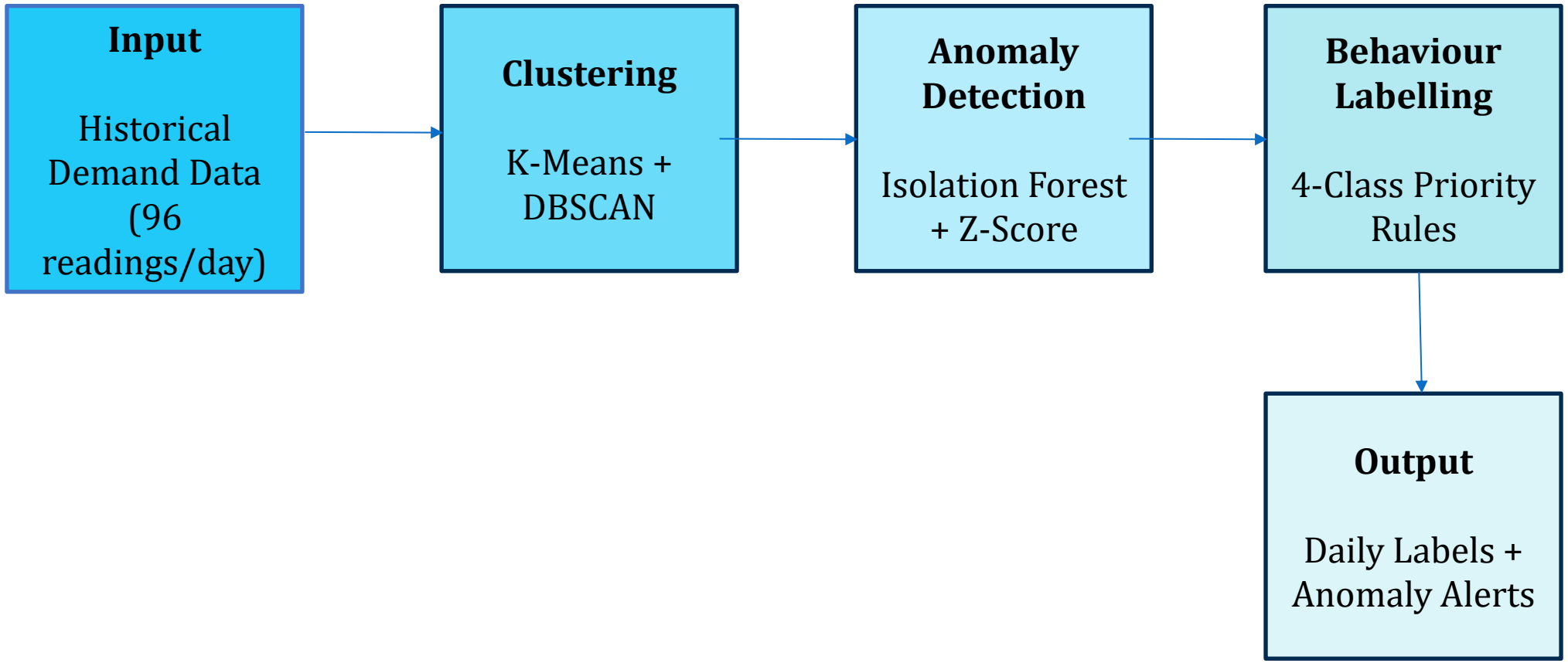
Component : Behaviour Intelligence

Comments: Domain knowledge in the proposed area seems to be insufficient

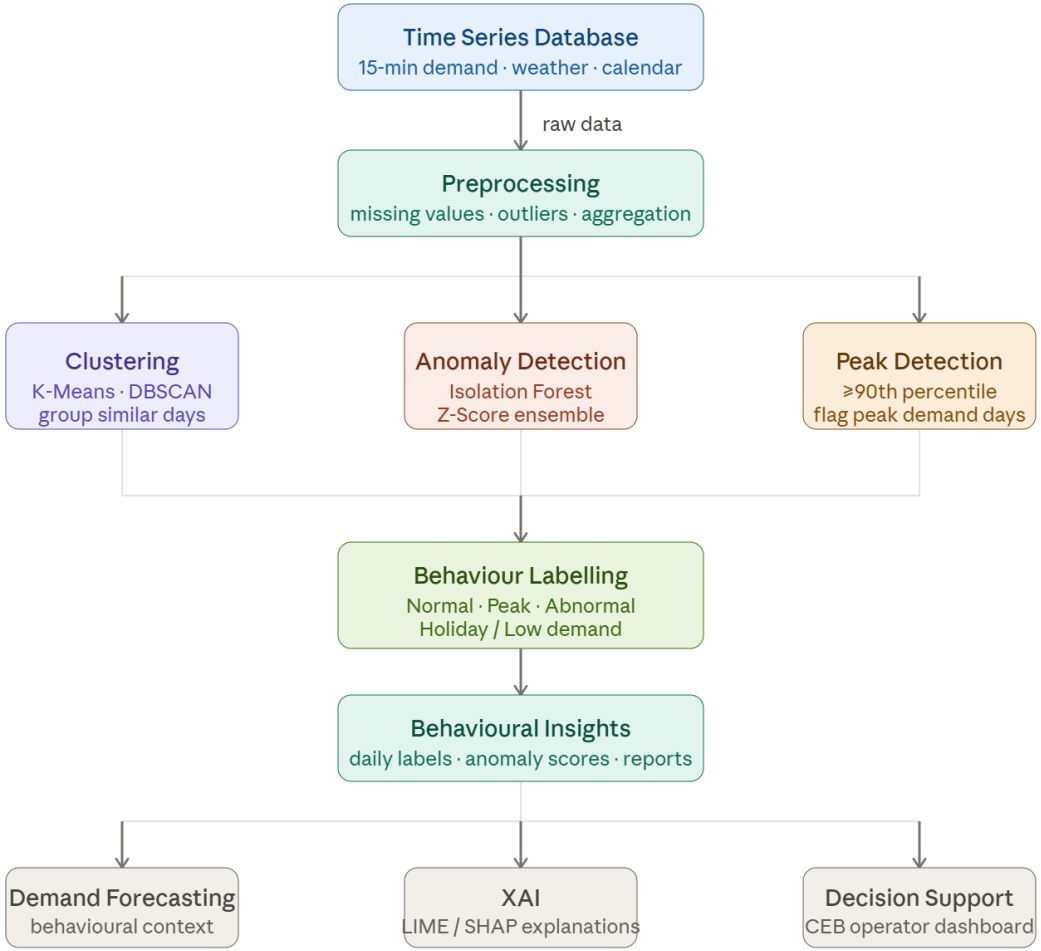
Purpose - Analyse daily electricity demand patterns by clustering similar days, detecting anomalous demand events, and generating interpretable behaviour labels for grid operators.

Key Differentiators
Day-level demand pattern clustering using K-Means + DBSCAN
Dual anomaly detection ensemble: Isolation Forest + Z-Score
4-class interpretable behaviour labelling: Normal / Peak / Abnormal / Low

Clear Component Pipeline



Component Architecture Diagram



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Behaviour Intelligence

Specialization: Information Technology

Behaviour Intelligence Module

The Sub Problem:

Existing forecasting systems produce only numerical demand predictions with no explanation of why demand behaves unusually on specific days. No automated system classifies each day as Normal, Peak, Abnormal, or Low demand - leaving CEB operators without actionable behavioural intelligence.

Our Solution:

- K-Means + DBSCAN Clustering
- Isolation Forest + Z-Score Anomaly Detection
- 4-Class Behaviour Labelling

Behaviour Intelligence Module

User Requirements

Energy Planners & Analysts

Daily behaviour labels and anomaly alerts to support generation scheduling and peak demand preparation

Grid Managers & Utility Staff

Monitor flagged anomalous demand days and make load balancing decisions with interpretable daily demand summaries

Policymakers & Researchers

Access transparent demand behaviour insights to inform long-term energy infrastructure planning and renewable integration strategies

Behaviour Intelligence Module

Design Excellence

Novelty

First unified framework for Sri Lanka combining day-level demand clustering, dual anomaly detection, and interpretable behaviour labelling in a single 5-stage pipeline, no existing system provides this.

Domain Depth

Validated on 189,888 CEB demand records (2020–2025). K-Means best K=6 with Silhouette Score 0.47. Dual ensemble (Isolation Forest + Z-Score) detects ~59 anomalous days. Evaluated using Davies-Bouldin Index, Precision, Recall, and F1-Score.

Practical Impact

Enables CEB operators to act on clear daily labels (Normal Weekday / Peak Demand / Holiday-Low / Abnormal) with XAI drill-down for anomalous days , solving the exact gap confirmed in panel feedback.

Behaviour Intelligence Module

Prototype



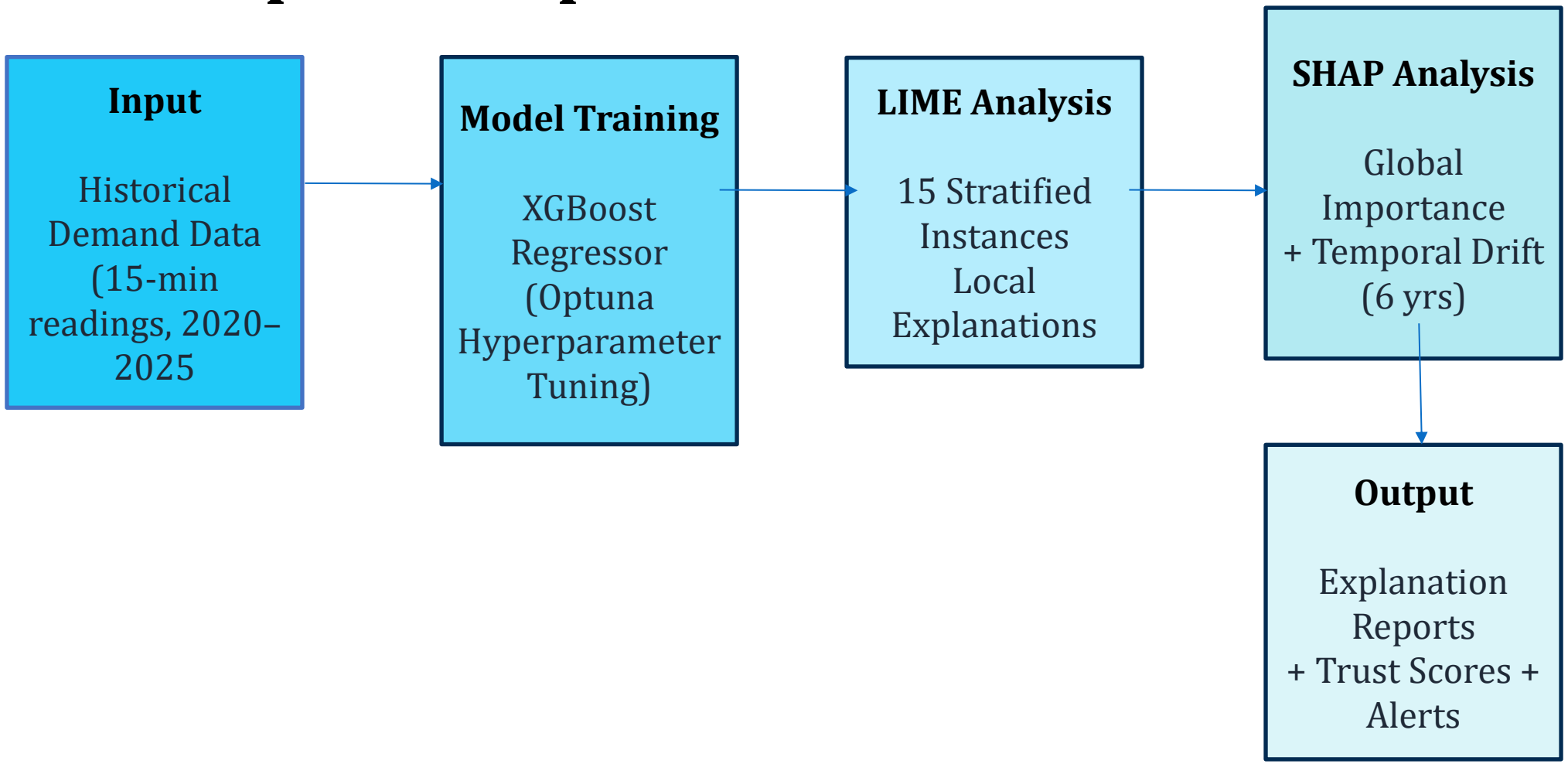
Component : XAI — LIME & SHAP Analysis

Comments: the components presented lack the depth for a 1 year research

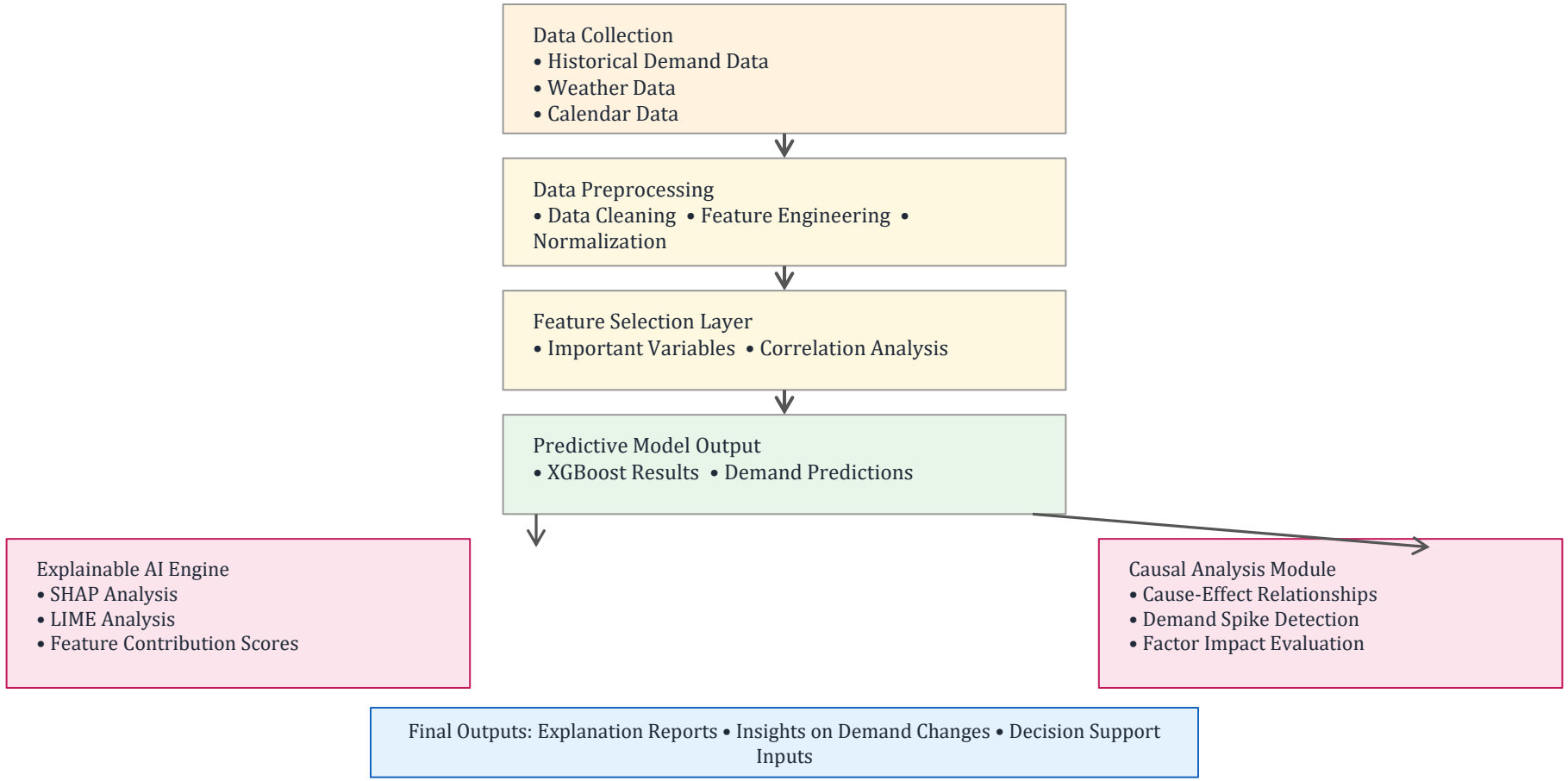
Purpose — Provide transparent, interpretable explanations for electricity demand forecasts using LIME (local instance-level explanations) and SHAP (global feature importance + temporal drift analysis), enabling CEB grid operators to understand WHY demand predictions are made.

Key Differentiators
Base model: XGBoost Regressor (R^2 = 0.9999) — production-grade accuracy ensures trustworthy XAI explanations
LIME — stratified 15-instance local explanation sampling across Low / Medium / High / Peak demand spectrum
SHAP — 6-year temporal drift analysis (2020–2025) tracking how feature importance shifts year over year

Clear Component Pipeline



Component Architecture Diagram



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IT22273062 | MADUSHANI R P

XAI: LIME & SHAP Analysis

Specialization: Information Technology

XAI: LIME & SHAP Analysis

The Sub Problem:

1. XGBoost and deep learning models are black boxes — they predict electricity demand accurately but cannot explain WHY demand behaves as it does on specific days.

2. Grid operators, regulators, and energy planners need justifications: "Why was demand predicted high on this Monday?" — no existing Sri Lankan system provides this.

3. Without explainability, CEB operators cannot trust AI predictions or act on them confidently during peak demand or anomalous events.

Our Solution:

XGBoost + Optuna Tuning	LIME Local Explanations	SHAP Temporal Drift Analysis
50-trial Bayesian optimisation (TPE sampler) → best hyperparameters → R^2 = 0.9999 base model with 0.14% RMSE/mean demand ratio	15 stratified instances across Low / Medium / High / Peak demand — per-instance linear approximation of XGBoost with 1,000 perturbations	Year-by-year (2020–2025) feature importance tracking — reveals how Temperature, Solar Irradiance, Hour of Day importance shifts over time

XAI: LIME & SHAP Analysis

User Requirements

Energy Planners & CEB Operators

Require transparent, feature-level explanations for daily demand forecasts — SHAP summary plots and LIME force charts that show exactly which features (temperature, hour, weekday) drove each prediction up or down, supporting peak demand preparation.

Grid Managers & Utility Staff

Need to interrogate anomalous demand days via XAI drill-down: when Behaviour Intelligence flags an anomaly, XAI auto-loads the LIME explanation for that specific date, enabling interpretable load balancing decisions with full feature contribution transparency.

Policymakers & Researchers

Access SHAP temporal drift reports (2020–2025) to understand how the importance of temperature, solar irradiance, and time-of-day has shifted over six years — enabling evidence-based long-term energy infrastructure and renewable integration strategies.

XAI: LIME & SHAP Analysis

Design Excellence

Research Novelty

First XAI framework for Sri Lanka combining LIME stratified sampling (15 instances across 4 demand tiers) with SHAP 6-year temporal drift analysis (2020–2025) — no existing system provides both local and temporal-global explainability for CEB demand forecasting in a single integrated pipeline.

Domain Depth

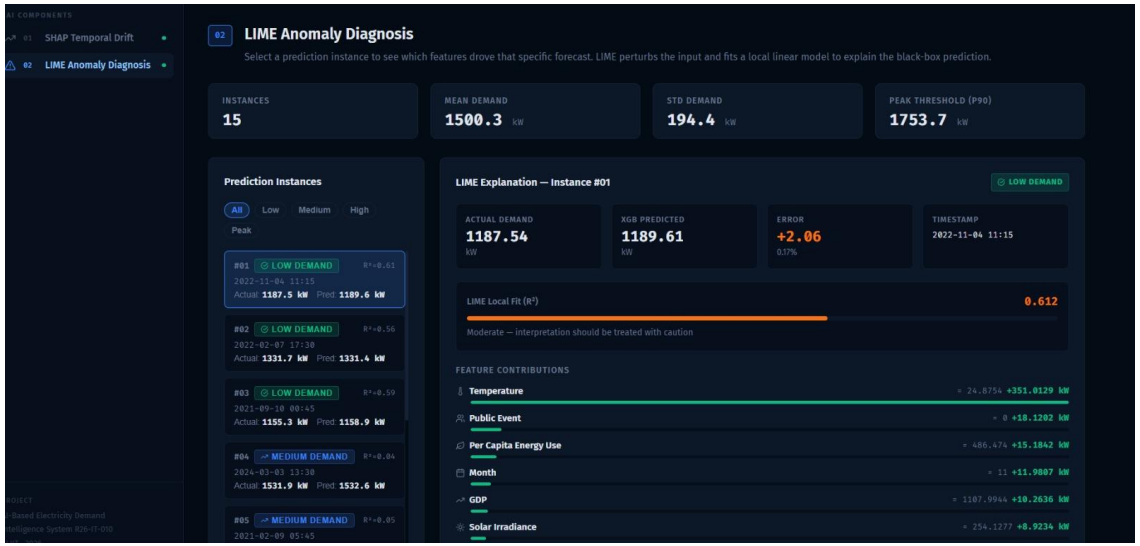
Validated on 188,607 CEB demand records (2020–2025). XGBoost base model: R^2 = 0.9999, RMSE = 2.03 kW, MAE = 1.05 kW (0.14% of mean demand 1,499.65 kW). Hyperparameter tuning: 50 Optuna trials (TPE sampler). SHAP drift: Temperature avg impact 157.74 kW, declining −2.67 kW over 6 years. LIME Trust Validator: R^2 threshold ≥ 0.90 = Excellent.

Practical Impact

Enables CEB operators to: (1) get per-prediction LIME explanations with trust scores, (2) view SHAP Demand Reduction Advisor for actionable vs non-actionable feature guidance, (3) receive Cohort Profiling for weekday/weekend demand pattern differences — directly integrated with Behaviour Intelligence anomaly detection via ?source=bi&date= API trigger.

XAI: LIME & SHAP Analysis

Prototype





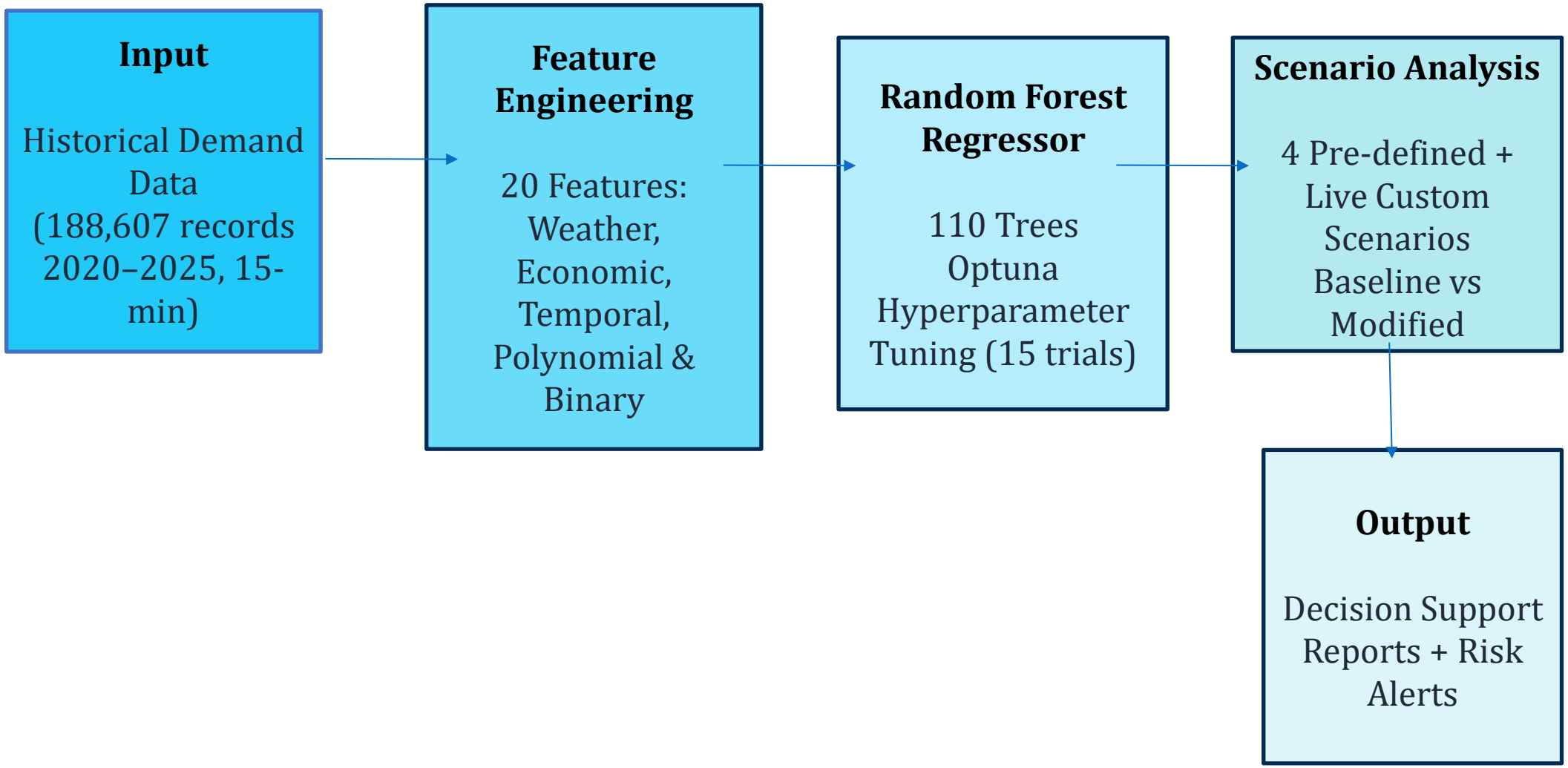
Component : Scenario Analysis & Decision Support System

Comments: component 3 & 4 appear to overlap

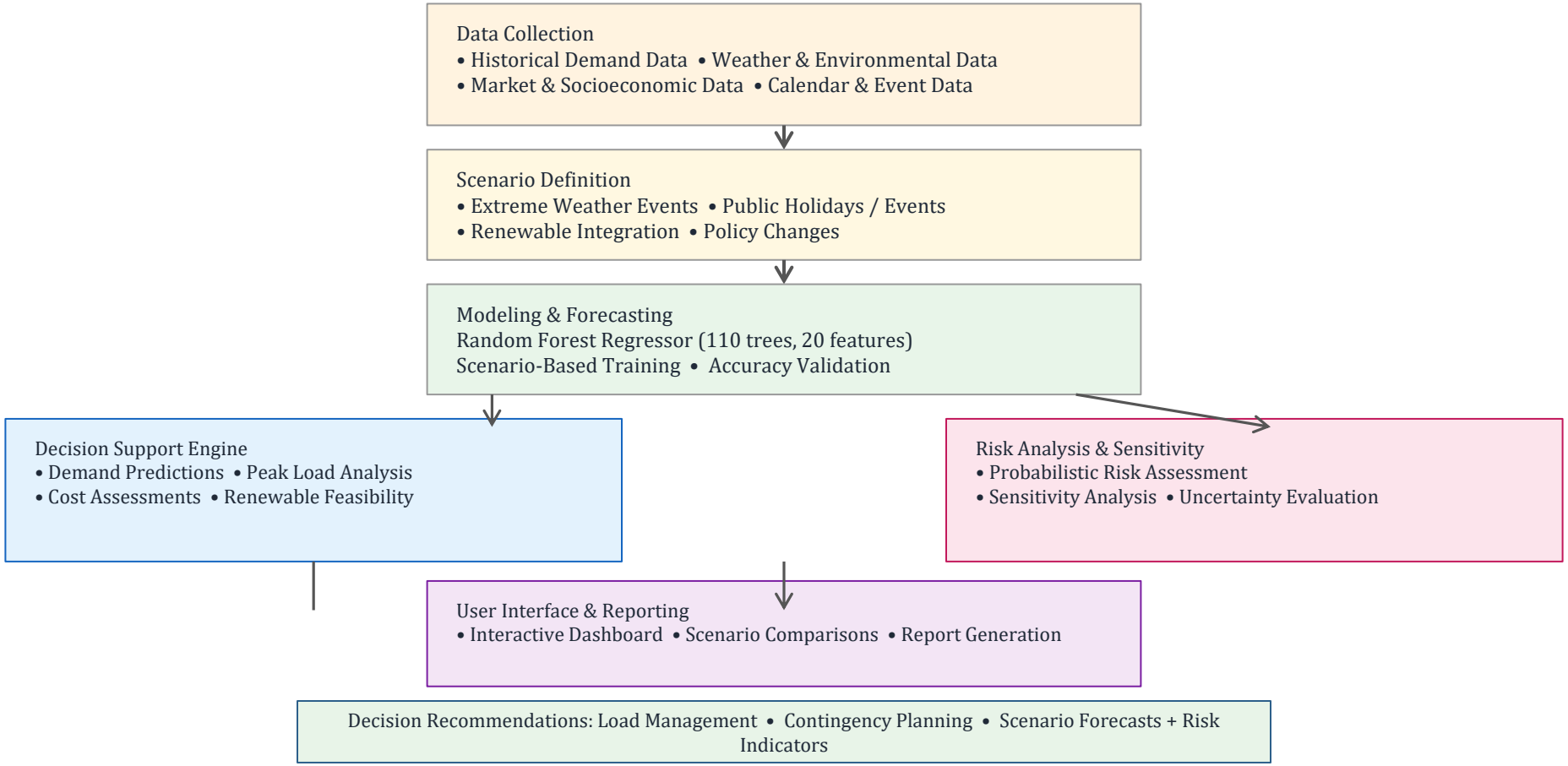
Purpose — Enable electricity planners, utility operators, and policymakers to simulate multiple 'what-if' demand scenarios, assess risks, and make informed data-driven decisions using a Random Forest forecasting model integrated with an interactive scenario analysis dashboard.

Key Differentiators
Random Forest Regressor — 110 trees, 20 engineered features — R^2 = 0.9998 (99.98%) on test set
4 pre-defined what-if scenario simulations: Heat Wave / Cold Spell / Price Shock / Economic Boom with quantified demand impact
Live custom scenario API — any baseline + any modification → instant kW change + % change response via /api/scenario-test

Clear Component Pipeline



Component Architecture Diagram



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Scenario Analysis & Decision Support System

Specialization: Information Technology

Scenario Analysis & Decision Support System

The Sub Problem:

1. Existing electricity forecasting systems produce only single-point demand predictions with no scenario comparison capability — planners cannot test 'what if' policy situations before implementing them.

2. Critical questions CEB operators cannot currently answer: 'If a heat wave hits, how much extra capacity do we need?' or 'If we double electricity prices, will demand fall?' — no interactive simulation tool exists for Sri Lanka.

3. Most systems lack explainable, risk-aware decision support — demand drivers remain opaque, making it difficult for planners to justify grid management decisions or prepare for sudden demand changes.

Our Solution:

Random Forest Regressor	4-Scenario What-If Engine	Live API + React Dashboard
110 trees, 20 engineered features — R^2 = 0.9998 on test set. Optuna Bayesian optimisation (15 trials, TPE sampler). Triple outlier removal: IQR + Z-score + Isolation Forest.	Pre-defined Heat Wave (+19.89%) / Cold Spell (−40.33%) / Price Shock (+0.002%) / Economic Boom (−0.17%) scenarios with quantified kW impact and policy implication.	FastAPI /api/scenario-test endpoint for custom baseline vs modified simulation. 4-tab React dashboard: Overview | Predict | Scenarios | Performance on port 5175.

Scenario Analysis & Decision Support System

User Requirements

Energy Planners & Analysts (Primary Users)

Run multiple 'what-if' scenarios — Heat Wave, Cold Spell, Price Shock, Economic Boom — to quantify demand impacts before they occur. Access the 4-tab React dashboard to visualise baseline vs scenario predictions, review all 20 feature contributions, and export results for generation scheduling decisions.

Utility Staff & Grid Managers (Secondary Users)

Use pre-computed scenario results and live /api/scenario-test endpoint to instantly simulate custom parameter modifications. Access risk indicators and auto-generated business insights per scenario (e.g. 'Heat waves require 20% extra grid capacity') to support load balancing and contingency planning.

Policymakers & Researchers (Tertiary Stakeholders)

Access the scenario engine's key research findings — temperature dominates 88% of demand decisions, demand is price-inelastic (0.002% change even if price doubles) — to inform long-term renewable energy integration strategies and evidence-based energy policy formulation for Sri Lanka.

Scenario Analysis & Decision Support System

Design Excellence

Research Novelty

First production-ready Sri Lanka electricity demand system combining Random Forest forecasting (R^2 = 0.9998) with live what-if scenario analysis via REST API — no existing local system allows planners to instantly quantify the demand impact of heat waves, price changes, or economic shifts through an interactive dashboard.

Domain Depth

Validated on 188,607 CEB records (2020–2025), 20 engineered features including polynomial (Temperature^2, IsHotDay) and interaction terms (TempHumidityInteraction). Key finding: Temperature-based features dominate at 88% combined importance. Price inelasticity confirmed: doubling electricity price causes only +0.002% demand change. Heat wave scenario: +19.89% demand, Cold spell: −40.33%.

Practical Impact

Enables CEB planners to act on quantified scenario insights: (1) Heat Wave planning — grid needs 20% extra capacity; (2) Cold spell response — 40% demand reduction; (3) Price policy — pricing alone cannot control peak demand; (4) Economic growth — GDP alone does not significantly increase residential grid demand. Live /api/scenario-test endpoint enables real-time custom simulation for any policy parameter.

Scenario Analysis & Decision Support System

Prototype

